Assignment 15

Chapter 11

In the following, $\mathbf{X}$, $\mathbf{Y}$, $\mathbf{Z}$, etc denote vectors of random variables.

5. Show that $R_{ii'}(0)$ in (11.234) vanishes if and only if $i \neq i'$.

8. Show that for a wide-sense stationary process $X(t)$, $S_X(f) = S_X(-f)$ for all f.

9. Consider a zero-mean white Gaussian noise process $Z(t)$. Let $h_1(t)$ and $h_2(t)$ be two impulse responses such that the supports of $H_1(f)$ and $H_2(f)$ do not overlap.

   (a) Show that for any t and t', the two random variables $Z(t) * h_1(t)$ and $Z(t) * h_2(t')$ are independent.

   (b) Show that the two processes $Z(t) * h_1(t)$ and $Z(t) * h_2(t)$ are independent.

   (c) Repeat a) and b) if $Z(t)$ is a zero-mean colored Gaussian noise process. Hint: Regard $Z(t)$ as obtained by passing a zero-mean white Gaussian noise process through a coloring filter.

10. Interpret the bandpass white Gaussian channel as a special case of the bandlimited colored Gaussian channel in terms of the channel capacity.

11. *Independent Gaussian noise is the worst* Let C be the capacity of a system of k Gaussian channels with $Z_i \sim \mathcal{N}(0, N_i)$. By ignoring the possible correlation among the noise variables, we can use the channels in the system independently as parallel Gaussian channels. Thus C is lower bounded by the expression in (11.180). In this sense, a Gaussian noise vector is the worst if its components are uncorrelated. Justify this claim analytically. Hint: Show that $I(\mathbf{X}; \mathbf{Y}) \geq \sum_i I(X_i; Y_i)$ if X_i are independent.

Supplementary Problems

1. (**Data processing inequality for informational divergence**) Let X, X', Y, and Y' be discrete random variables with $\mathcal{X} = \mathcal{X}'$ and $\mathcal{Y} = \mathcal{Y}'$, such that $p_{XY}(x, y) = p_X(x)w(y|x)$ and $p_{X'Y'}(x, y) = p_{X'}(x)w(y|x)$, where $w(\cdot|\cdot)$ is a transition matrix from $\mathcal{X}$ to $\mathcal{Y}$.

   (a) Prove that $D(p_{XY} \| p_{X'Y'} = D(p_X \| p_{X'})$.

   (b) Prove that $D(p_X \| p_{X'}) \geq D(p_Y \| p_{Y'})$. Hint: Prove that $D(p_{XY} \| p_{X'Y'}) \geq D(p_Y \| p_{Y'})$.

2. Using the result in Problem 1, prove that if $X \to Y \to Z$, then $I(X; Z) \leq I(X; Y)$ (Lemma 2.41).